

Gel electrophoresis (Southern)

Preparation

Material	Concentration	Final concentration	Location
PCR product	--	--	Store at 4°C
TAE buffer ※	50X	0.5X	II
Agarose gel †	--	3%	II
SYBR® stock solution *	1000X	1X	II
Loading Dye	6X	1X	4°C (c)
RTU-100 (Ladder marker)	--	--	4°C (c)

※ To prepare the 0.5X TAE buffer, make at least 1 L, as each run requires at least 500 mL 0.5X TAE buffer;

† The big gel slot requires around 40 mL gel solution; the small gel slot requires 20 mL gel solution. For example, to make a 3% big gel, add 1.2 g agarose powder (ultra pure grade) to 40 mL 0.5X TAE buffer.

* To prepare 1000X SYBR stock solution, dilutes 100 µL SYBR® raw solution with 900 µL DMSO. This chemical is used to stain the DNA and detected by UV radiation.

Procedure

Step 1: Make gel

1. Prepare the gel. After adding agarose powder into the 0.5X TAE buffer, melt them with microwave. Gently shake the vial during each time points.

- use mid-low firepower, 1 mins → 30 secs → add ddH₂O → 20 secs → 10 secs

2. Add 0.001 volume of SYBR® stock solution into the gel solution immediately (1000X → 1X), gently shake to mix them well;

- For example, 40 mL gel requires 40 µL 1000X SYBR stock solution.

3. Pour the gel into the gel container and wait for solidification (around 15 mins);

Step 2: Set up the electrode machine

4. Prepare the electrophoresis machine, the electrode direction should be from negative to positive;
5. After the gel was solidified, put the gel container together with the gel into the electrophoresis machine;
6. Add some TAE buffer (same strength with the gel; 0.5X in this case) into the electrophoresis machine and make sure the gel was fully submerged in the buffer;

Step 3: Load sample

7. Add loading dye to the PCR product. The loading dye final concentration should be 1X.
For example, 2 μ L 6X loading dye + 10 μ L PCR product;
8. Gently inject 2 μ L ladder marker into the first and the last wells of the gel;
9. Gently inject the blank and the PCR products (around 6 μ L) into the wells;

Step 4: Start running

10. Set the voltage as **135 V** and the runtime as **40 mins**;

5 ~ 10 V / cm for DNA size < 1 kb;

4 ~ 10 V / cm for DNA size between 1 ~ 12 kb;

1 ~ 3 V / cm for DNA size greater than 12 kb.

For our machine, the distance between the electrodes is around 13.5 cm.

Step 5: Gel-imaging

1. Turn on the machine (Bio-Rad Gel Doc XR+);
2. Put in the gel;
3. Start up the "Image Lab" software;
4. New protocol;
5. Gel imaging;
6. Application → Select → Nucleic acid → SYBR[®] Safe;
7. Position Gel → Filter 1 → adjust gel position;
8. Run protocol;
9. Export for publication (Export for analysis is 16-bit images).

Done !!!
